ML-Based Weekly Stock Ranking & Portfolio Allocation

Algorithm Architects

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Problem & Stakeholder

THE PROBLEM

College students who just started investing pick stocks based on Reddit posts, TikTok tips, or what friends say. They have no financial background, no systematic approach, and make emotional decisions that often lead to losses. They need a simple, data-driven way to decide what to buy and what to avoid each week.

WHO

College students aged 18–25 who recently opened a Robinhood or Webull account with $500–2,000 in savings. No finance background. Currently making trades based on social media hype and gut feeling.

WHAT THEY NEED

A simple weekly signal: which stock is most likely to go up, which to avoid. Not a complex dashboard — just a clear ranking to check once a week before making trades. Replace gut decisions with data.

Envisioned Solution

Vhat we propose to build for our stakeholder

he Idea

A student enters how much money they have — say $500. The system runs 6 ML models on 15 stocks, predicts which ones are most likely to go up next week, and tells them: "Put $200 in GS (72% confidence), $150 in CVX (65% onfidence), $100 in JPM (61% confidence), keep $50 as cash." One simple screen, once a week, no finance degree eeded.

Predict

models analyze 22 features from he past 4 weeks and predict UP or OWN probability for each stock.

Rank & Allocate

Stocks are ranked by confidence. Higher ranked stocks get a larger share of the student's budget. Lower ranked ones get less or nothing.

Reduce Risk

When multiple models agree, confidence is high. When they disagree, the system flags uncertainty and suggests keeping cash instead of risking it.

places gut feeling and social media hype with a data-driven weekly investment plan tailored to the student's budget.

Data

Source & Structure

Attribute	Details
Source	Yahoo Finance — daily OHLCV data downloaded as CSV
Stocks (Core)	AAPL, MSFT, GOOGL, AMZN, NVDA (5 tech stocks)
Stocks (Expanded)	+META, TSLA, JPM, GS, JNJ, PFE, XOM, CVX, WMT, BA (15 total)
Date Range	May 2015 – Feb 2026 (~564 weeks per stock)
Raw Columns	6 (Date, Open, High, Low, Close, Volume) — all quantitative
Engineered Features	22 per stock (returns, RSI, MACD, Bollinger, volatility, volume ratio, cross-sectional ranks)
Frequency	Resampled from daily to weekly (Friday close)
Train / Val / Test	Before 2022 / 2022–2023 / 2024+ (time-based, no data leakage)

Data Quality Validation

Checked for missing values (dropna after each step) · Coerced non-numeric entries (pd.to_numeric errors='coerce') · Stripped whitespace from column names · Verified chronological ordering (sort by Date) · Confirmed no duplicate dates · Validated Volume > 0 · Verified class balance (~57–61% UP weeks, no severe imbalance)

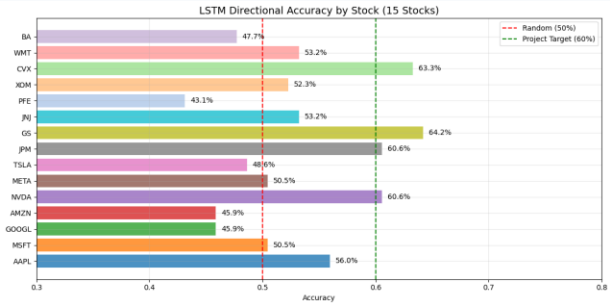
Initial Results

Model	Accuracy	F1 Score	Stocks	Best Stock
LSTM	43–64%	0.63	5&15	NVDA (59.6%) & GS (64.2%)
Random Forest	41–59%	0.56	15	AMZN (59.3%)
ARIMA	46–61%	0.47–0.66	5	AMZN (61.4%)
Gradient Boost.	53.7%	0.64	15	GOOGL (60.69%)
XGBoost	38–59%	0.00–0.74	5	NVDA (59.2%)
Elastic Net	57.4%	0.73	15	JNJ (69.5%)

Key Findings

- All models beat the 50% random baseline
- NVDA best predicted stock across models (59.6% LSTM, 59.2% XGBoost). AI demand created a learnable trend
- Results consistent with financial ML literature — pro quant funds achieve 52–55% directional accuracy
- LSTM selected as best overall model: highest F1 (0.63), predicts both UP and DOWN

Initial Results — LSTM(15 Stocks) & Backtest



Best Model: LSTM

Architecture: LSTM(64)→LSTM(32)→Dense(1)
Input: 4 weeks × 22 features per stock
Output: P(UP) — probability stock rises
Overall F1: 0.63 | Top-1: 12.8% (random: 6.7%)
4 stocks crossed 60% target: GS, CVX, JPM, NVDA

What Worked

- 4 stocks crossed 60% target — GS (64.2%), CVX (63.3%), JPM (60.6%), NVDA (60.6%)
- Top-1 ranking accuracy 12.8% vs 6.7% random — nearly double
- Top-2 accuracy 25.7% vs 13.3% random

What Didn't

- Elastic Net always predicted BUY — lost 51% of portfolio
- PFE (43.1%), GOOGL (45.9%) — worse than random for some stocks
- Per-stock F1 dropped from 0.70 (5 stocks) to 0.63 (15 stocks) due to sector diversity

Challenges

Narrow probability spread

LSTM predicts UP for nearly all weeks on some stocks (MSFT: 109/109 up, AMZN: 109/109 up). Probability std as low as 0.017 — too narrow to differentiate stocks for ranking.

5 vs 15 stock mismatch

ARIMA, XGBoost use 5 tech stocks while LSTM, RF, GB, Elastic Net use 15 across sectors. Makes direct ranking comparison imperfect until all models use the same stock universe.

Regression vs classification mismatch

XGBoost uses regression (predicts exact return) while others use classification. Causes extreme F1 variation — AMZN F1 = 0.00 because predictions cluster near zero.

Market regime changes

Patterns learned in 2015–2021 don't always hold in 2024–2026. This is inherent to financial ML, not a bug in the model.

Classification bias toward UP

Markets trend upward over time (~57–61% of weeks are UP). Models can achieve decent accuracy by just predicting UP always, without learning real signal.

Plan & Goals

Final deliverable goals

1. **Expand LSTM, XGBoost and ARIMA to 15 stocks**
   Currently use 5 tech stocks — expanding to 15 across sectors enables fair cross-model ranking comparison

2. **Improve probability spread**
   Add richer features (sentiment, earnings dates, sector momentum) to widen LSTM probability distribution for better stock differentiation

3. **Ensemble ranking strategy**
   Combine predictions from all 6 models into a majority-vote or weighted-average ranking system

4. **Full backtest with benchmarks**
   Compare portfolio returns against S&P 500, Nasdaq, and equal-weight baseline over the same test period

5. **Final report**
   Document methodology, results, and limitations with complete reproducible code on GitHub